LOGISTICS DELIVERY PERFORMANCE & COST ANALYTICS

Complete Beginner-to-Intermediate Data Analytics Assignment Documentation

Project Type: Business Analytics + Data Analytics

Difficulty: Beginner → Intermediate

Domain: Logistics / Delivery Operations

Primary Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Streamlit

Development Style: Modular Python + Data Cleaning Pipeline

Machine Learning: Not included

Assignment Style: Standardized implementation — all students must use the same folder structure, file names, function names, and required outputs.

---

1. PROJECT OVERVIEW

1.1 Project Title

Logistics Delivery Performance & Cost Analytics

1.2 Project Scenario

You are working as a **Data Analyst / Business Analyst** for a logistics company.

The company manages customer orders, shipments, warehouses, transportation, delivery partners, delivery dates, and logistics costs.

Management currently has difficulty answering important business questions:

- How many orders are delivered on time?
- How many orders are delayed?
- Which routes experience more delays?
- Which warehouses take longer to process shipments?
- Which transportation modes are expensive?
- Which delivery partners perform differently?
- What is the average logistics cost?
- Which areas require operational attention?
- What business actions can improve delivery performance and cost control?

The company wants a data analytics solution that converts raw operational data into meaningful business information.

---

2. BUSINESS OBJECTIVE

The main objective is:

Analyze logistics operational data, identify delivery and cost patterns, calculate business KPIs, create useful visualizations, and provide clear business insights and recommendations.

The project must focus on:

```
Raw Data
   ↓
Data Understanding
   ↓
Data Cleaning
   ↓
Cleaning Pipeline
   ↓
EDA
   ↓
Business Analysis
   ↓
KPI Analysis
   ↓
Visualization
   ↓
Business Insights
   ↓
Recommendations
   ↓
Streamlit Dashboard
```

---

3. PROJECT SCOPE

3.1 In Scope

The project must include:

- 1. Business understanding
- 2. BRD
- 3. FRD
- 4. Dataset understanding
- 5. Data quality analysis
- 6. Data cleaning
- 7. Reusable data cleaning pipeline
- 8. Exploratory Data Analysis
- 9. Business Analytics
- 10. KPI calculation
- 11. Statistical analysis
- 12. Matplotlib visualizations
- 13. Seaborn visualizations
- 14. Business insights
- 15. Business recommendations
- 16. Streamlit dashboard
- 17. Testing
- 18. Final project documentation

3.2 Out of Scope

The project must NOT include:

- Machine Learning

- Deep Learning
- NLP
- Generative AI
- RAG
- MLflow
- Docker
- Kubernetes
- Model deployment
- Prediction models
- Complex databases
- Cloud infrastructure

---

4. TECHNOLOGY STACK

4.1 Programming Language

Python

4.2 Data Analysis

- Pandas
- NumPy

4.3 Visualization

- Matplotlib
- Seaborn

4.4 Development

- Jupyter Notebook
- VS Code / PyCharm / similar IDE

4.5 Dashboard

Streamlit

4.6 Data Format

CSV

---

5. STANDARD PROJECT FOLDER STRUCTURE

Every student MUST use this structure.

```
logistics_delivery_analytics/
```

```
■
■■■ data/
■   ■■■ raw/
■   ■   ■■■ logistics_data.csv
■   ■
■   ■■■ cleaned/
■       ■■■ logistics_cleaned.csv
■
■■■ notebooks/
■   ■■■ logistics_analysis.ipynb
■
■■■ src/
■   ■■■ __init__.py
■   ■■■ data_loader.py
■   ■■■ data_cleaner.py
■   ■■■ business_analysis.py
■   ■■■ visualization.py
■
■■■ app/
■   ■■■ streamlit_app.py
■
■■■ reports/
■   ■■■ business_report.md
■
■■■ tests/
■   ■■■ test_pipeline.py
■
■■■ run_pipeline.py
■■■ requirements.txt
■■■ README.md
■■■ .gitignore
```

5.1 Important Assignment Rule

Students must NOT rename the following:

```
data_loader.py
data_cleaner.py
business_analysis.py
visualization.py
streamlit_app.py
run_pipeline.py
logistics_analysis.ipynb
logistics_data.csv
logistics_cleaned.csv
business_report.md
test_pipeline.py
```

Students must also use the exact required function names defined in this document.

They may add comments and explanatory markdown, but they should not replace the standard functions with different names.

---

6. PURPOSE OF EACH FOLDER

6.1 data/raw

Contains the original client dataset.

```
data/raw/logistics_data.csv
```

Rule:

Never directly modify the raw file.

6.2 data/cleaned

Contains the final cleaned dataset.

```
data/cleaned/logistics_cleaned.csv
```

This file is produced by the cleaning pipeline.

6.3 notebooks

Contains the main analysis notebook.

```
notebooks/logistics_analysis.ipynb
```

The notebook demonstrates the complete analytical process.

6.4 src

Contains reusable Python functions.

```
src/
■■■ data_loader.py
■■■ data_cleaner.py
■■■ business_analysis.py
■■■ visualization.py
```

The purpose is to avoid putting all code into one large Python file.

6.5 app

Contains the Streamlit application.

```
app/streamlit_app.py
```

6.6 reports

Contains the final business report.

```
reports/business_report.md
```

6.7 tests

Contains basic validation tests.

```
tests/test_pipeline.py
```

Testing is kept simple and beginner-friendly.

---

7. DATASET DESIGN

The standard project can use one primary CSV file.

```
logistics_data.csv
```

The dataset should contain logistics order/shipment records.

7.1 Required Columns

Students must work with these standard column names:

```
order_id
customer_id
order_date
warehouse
origin_city
destination_city
distance_km
product_category
quantity
weight_kg
shipping_mode
delivery_partner
shipping_cost
fuel_cost
warehouse_processing_hours
dispatch_date
expected_delivery_date
actual_delivery_date
delivery_status
customer_rating
damage_flag
return_flag
```

---

8. DATA DICTIONARY

Column	Meaning	Type
order_id	Unique order identifier	String
customer_id	Customer identifier	String
order_date	Date order was created	Date
warehouse	Warehouse handling the order	String
origin_city	Shipment starting city	String
destination_city	Delivery city	String
distance_km	Delivery distance	Numeric
product_category	Product category	String
quantity	Number of products/items	Integer
weight_kg	Shipment weight	Numeric
shipping_mode	Transportation method	String
delivery_partner	Delivery company/partner	String
shipping_cost	Shipment transportation cost	Numeric
fuel_cost	Fuel-related cost	Numeric
warehouse_processing_hours	Warehouse processing duration	Numeric
dispatch_date	Shipment dispatch date	Date
expected_delivery_date	Expected delivery date	Date
actual_delivery_date	Actual delivery date	Date
delivery_status	Delivered/Delayed/Cancelled	String
customer_rating	Customer rating	Numeric
damage_flag	Whether shipment was damaged	0/1
return_flag	Whether shipment was returned	0/1

---

9. DERIVED COLUMNS

The analyst must create the following derived columns.

9.1 Delivery Days

```
delivery_days =
actual_delivery_date - dispatch_date
```

9.2 Delay Days

```
delay_days =
actual_delivery_date - expected_delivery_date
```

For business reporting:

```
delay_days <= 0  → On Time
delay_days > 0   → Delayed
```

9.3 Cost Per KM

```
cost_per_km =
shipping_cost / distance_km
```

Handle zero distance safely.

9.4 Total Logistics Cost

```
total_logistics_cost =
shipping_cost + fuel_cost
```

9.5 On-Time Flag

```
on_time_flag =
1 if delay_days <= 0
0 if delay_days > 0
```

---

10. BUSINESS REQUIREMENTS DOCUMENT (BRD)

10.1 Business Problem

The logistics company has operational data but does not have a clear analytical view of delivery performance, delays, route efficiency, warehouse operations, transportation costs, and delivery partner performance.

Management needs a reliable analytics solution.

---

11. BUSINESS OBJECTIVES

The project must help management:

- 1. Measure delivery performance.
- 2. Identify delayed shipments.
- 3. Understand delay patterns.
- 4. Analyze routes.
- 5. Compare warehouse performance.
- 6. Analyze transportation modes.
- 7. Analyze delivery partners.

- 8. Understand logistics costs.
- 9. Monitor customer experience indicators.
- 10. Make data-supported operational decisions.

---

12. BUSINESS QUESTIONS

12.1 Delivery Questions

- 1. How many total orders exist?
- 2. How many orders were delivered?
- 3. How many orders were delayed?
- 4. What percentage of orders were delivered on time?
- 5. What is the average delivery time?
- 6. What is the average delay?

12.2 Route Questions

- 7. Which routes handle the most shipments?
- 8. Which routes have the highest average delay?
- 9. Which routes have the highest delivery cost?
- 10. Which routes have the highest cost per kilometer?

12.3 Warehouse Questions

- 11. Which warehouse processes the most orders?
- 12. Which warehouse has the highest average processing time?
- 13. Which warehouse has the highest delay percentage?

12.4 Transportation Questions

- 14. Which shipping mode is used most frequently?
- 15. Which shipping mode has the highest average cost?
- 16. Which shipping mode has the highest average delivery time?
- 17. Which shipping mode has the highest delay percentage?

12.5 Delivery Partner Questions

- 18. Which partner handles the most shipments?
- 19. Which partner has the highest on-time percentage?
- 20. Which partner has the highest average shipping cost?
- 21. Which partner has the highest damage rate?

12.6 Cost Questions

- 22. What is the total logistics cost?
- 23. What is the average logistics cost per order?

- 24. What is the average shipping cost?
- 25. What is the total fuel cost?
- 26. How does cost change by route, warehouse, mode, and partner?

12.7 Customer Questions

- 27. What is the average customer rating?
- 28. Is customer rating different across delivery status?
- 29. What percentage of shipments are returned?
- 30. What percentage of shipments are damaged?

---

13. BUSINESS OUTCOMES

The final project should provide:

```
Reliable cleaned data
        +
Clear KPIs
        +
Business analysis
        +
Visual evidence
        +
Business insights
        +
Actionable recommendations
```

The final answer must not simply say:

"Here is a graph."

It should explain:

What happened → Why it matters → What the business should investigate or improve.

---

14. FUNCTIONAL REQUIREMENTS DOCUMENT (FRD)

FR-01: Data Loading

The system must load the raw CSV dataset.

Required function:

```
load_data(file_path)
```

File:

```
src/data_loader.py
```

FR-02: Data Profiling

The analyst must inspect:

- Shape
- Columns
- Data types
- Missing values
- Duplicate rows
- Unique values
- Numerical summary
- Categorical summary

This can initially be performed in the notebook.

FR-03: Data Cleaning

The system must clean the dataset.

Required function:

```
clean_data(df)
```

File:

```
src/data_cleaner.py
```

FR-04: Missing Value Handling

Required function:

```
handle_missing_values(df)
```

The function must handle missing values according to the defined business rules.

FR-05: Duplicate Handling

Required function:

```
remove_duplicates(df)
```

Duplicate records must be identified and removed according to the business rules.

FR-06: Text Cleaning

Required function:

```
clean_text_columns(df)
```

Examples:

- Remove unnecessary spaces
- Standardize text case

• Handle inconsistent text values

FR-07: Date Cleaning

Required function:

```
clean_date_columns(df)
```

Required date columns:

```
order_date
dispatch_date
expected_delivery_date
actual_delivery_date
```

FR-08: Numeric Validation

Required function:

```
validate_numeric_columns(df)
```

Check:

```
distance_km
quantity
weight_kg
shipping_cost
fuel_cost
warehouse_processing_hours
customer_rating
```

FR-09: Business Rule Validation

Required function:

```
validate_business_rules(df)
```

FR-10: Feature/Derived Column Creation

Required function:

```
create_derived_columns(df)
```

It must create:

```
delivery_days
delay_days
cost_per_km
total_logistics_cost
on_time_flag
```

FR-11: Save Clean Data

Required function:

```
save_cleaned_data(df, output_path)
```

Output:

```
data/cleaned/logistics_cleaned.csv
```

---

15. STANDARD DATA CLEANING FUNCTION FLOW

The required flow is:

```
load_data()
      ↓
clean_data()
      ↓
handle_missing_values()
      ↓
remove_duplicates()
      ↓
clean_text_columns()
      ↓
clean_date_columns()
      ↓
validate_numeric_columns()
      ↓
validate_business_rules()
      ↓
create_derived_columns()
      ↓
save_cleaned_data()
```

Students may call the functions internally from `clean_data()`.

---

16. BUSINESS RULES

BR-01 Order ID

`order_id` should not be missing.

BR-02 Customer ID

`customer_id` should not be missing.

BR-03 Distance

```
distance_km >= 0
```

BR-04 Quantity

```
quantity > 0
```

BR-05 Weight

```
weight_kg >= 0
```

BR-06 Shipping Cost

```
shipping_cost >= 0
```

BR-07 Fuel Cost

```
fuel_cost >= 0
```

BR-08 Customer Rating

Assuming a 1–5 rating:

```
1 <= customer_rating <= 5
```

BR-09 Dates

Expected delivery date should normally not occur before dispatch date.

Actual delivery date should be checked for logical consistency.

BR-10 Delay

```
delay_days > 0 → Delayed
delay_days <= 0 → On Time
```

BR-11 Cost Per KM

Do not divide by zero.

---

17. DATA CLEANING PIPELINE

Pipeline Objective

Convert:

```
Raw Client Data
      ↓
Validated Data
      ↓
Cleaned Data
      ↓
Business-Ready Data
```

Pipeline Entry Point

File:

```
run_pipeline.py
```

Required function:

```
run_pipeline()
```

Expected execution:

```
python run_pipeline.py
```

Expected output:

```
Loading raw data...
Checking data quality...
Cleaning data...
Applying business rules...
Creating derived columns...
Saving cleaned data...
Pipeline completed successfully.
```

---

18. REQUIRED `data_loader.py` FUNCTIONS

```
load_data(file_path)
```

Purpose:

Load CSV data into a Pandas DataFrame.

Optional helper:

```
check_file_exists(file_path)
```

Students should keep the implementation simple.

---

19. REQUIRED `data_cleaner.py` FUNCTIONS

The standard functions are:

```
handle_missing_values(df)
remove_duplicates(df)
clean_text_columns(df)
clean_date_columns(df)
validate_numeric_columns(df)
validate_business_rules(df)
create_derived_columns(df)
clean_data(df)
save_cleaned_data(df, output_path)
```

Every function must have a docstring.

Example:

```
def remove_duplicates(df):
    """Remove duplicate records from the logistics DataFrame."""
```

---

20. BUSINESS ANALYSIS FUNCTIONS

File:

```
src/business_analysis.py
```

Students MUST use these function names.

Basic KPIs

```
calculate_total_orders(df)
calculate_total_delivered(df)
calculate_total_delayed(df)
calculate_on_time_percentage(df)
calculate_average_delivery_days(df)
calculate_average_delay_days(df)
```

Cost KPIs

```
calculate_total_shipping_cost(df)
calculate_average_shipping_cost(df)
calculate_total_fuel_cost(df)
calculate_total_logistics_cost(df)
calculate_average_logistics_cost(df)
calculate_average_cost_per_km(df)
```

Route Analysis

```
analyze_route_performance(df)
analyze_route_cost(df)
analyze_route_delays(df)
```

Warehouse Analysis

```
analyze_warehouse_performance(df)
analyze_warehouse_delays(df)
analyze_warehouse_processing_time(df)
```

Shipping Analysis

```
analyze_shipping_mode(df)
analyze_shipping_mode_cost(df)
analyze_shipping_mode_delays(df)
```

Partner Analysis

```
analyze_delivery_partner(df)
analyze_partner_cost(df)
analyze_partner_delays(df)
analyze_partner_damage_rate(df)
```

Customer Analysis

```
calculate_average_customer_rating(df)
calculate_return_rate(df)
calculate_damage_rate(df)
```

---

21. KPI FUNCTION OUTPUT

The functions should return simple Pandas objects or numeric values.

Example:

```
calculate_total_orders(df)
        ↓
Integer
analyze_route_performance(df)
```

```
        ↓
DataFrame

calculate_on_time_percentage(df)
        ↓
Float
```

Avoid unnecessary custom classes or complicated objects.

---

22. KPI DEFINITIONS

Total Orders

```
Number of unique orders
```

Total Delivered

```
Number of delivered orders
```

Total Delayed

```
Number of delayed orders
```

On-Time Delivery %

```
On-Time Deliveries / Delivered Orders × 100
```

Average Delivery Days

```
Average(actual_delivery_date - dispatch_date)
```

Average Delay Days

```
Average(delay_days for delayed shipments)
```

Average Shipping Cost

```
Total shipping cost / number of orders
```

Total Logistics Cost

```
shipping_cost + fuel_cost
```

Cost Per KM

```
shipping_cost / distance_km
```

Damage Rate

```
Damaged Shipments / Total Shipments × 100
```

Return Rate

```
Returned Shipments / Total Shipments × 100
```

---

23. EDA REQUIREMENTS

The notebook must contain three major EDA levels.

23.1 Univariate Analysis

Analyze one variable at a time.

Examples:

```
distance_km
weight_kg
shipping_cost
customer_rating
delivery_status
shipping_mode
warehouse
```

Recommended charts:

```
Histogram
Box Plot
Count Plot
Bar Chart
```

---

24. BIVARIATE ANALYSIS

Analyze relationships between two variables.

Examples:

```
Distance vs Delivery Days

Shipping Cost vs Distance

Warehouse vs Delay

Shipping Mode vs Cost

Partner vs Delivery Time

Delivery Status vs Customer Rating
```

Recommended charts:

```
Scatter Plot
Bar Plot
Box Plot
Count Plot
```

---

25. MULTIVARIATE ANALYSIS

Use multiple variables when useful.

Examples:

```
Warehouse + Shipping Mode + Delay

Partner + Cost + Delivery Status

Route + Cost + Delivery Time
```

Do not make unnecessarily complicated visualizations.

The goal is business understanding.

---

26. REQUIRED VISUALIZATIONS

Students must create at least:

Delivery

- 1. Delivery status distribution
- 2. Monthly order trend
- 3. Monthly delivery performance

Delay

- 4. Delay by warehouse
- 5. Delay by shipping mode
- 6. Delay by route
- 7. Delay distribution

Cost

- 8. Cost by shipping mode
- 9. Cost by route
- 10. Cost trend

Warehouse

- 11. Orders by warehouse
- 12. Processing time by warehouse

Partner

- 13. Shipments by partner
- 14. Partner delivery performance
- 15. Partner cost comparison

Customer

- 16. Customer rating distribution
- 17. Rating by delivery status

---

27. VISUALIZATION FUNCTIONS

File:

```
src/visualization.py
```

Required functions:

```
plot_delivery_status(df)
plot_monthly_orders(df)
plot_monthly_delivery_performance(df)
plot_delay_by_warehouse(df)
plot_delay_by_shipping_mode(df)
plot_delay_by_route(df)
plot_delay_distribution(df)
plot_cost_by_shipping_mode(df)
plot_cost_by_route(df)
plot_cost_trend(df)
plot_orders_by_warehouse(df)
plot_processing_time_by_warehouse(df)
plot_shipments_by_partner(df)
plot_partner_delivery_performance(df)
plot_partner_cost(df)
plot_customer_rating_distribution(df)
plot_rating_by_delivery_status(df)
```

Each function should:

- 1. Receive a DataFrame.
- 2. Perform only the required aggregation.
- 3. Create the chart.
- 4. Add a meaningful title.
- 5. Label axes.
- 6. Display or return the chart.

---

28. JUPYTER NOTEBOOK STRUCTURE

File:

```
notebooks/logistics_analysis.ipynb
```

The notebook must follow this exact order.

Section 1 — Business Understanding

Explain:

- Client
- Business problem
- Objectives
- Business questions

Section 2 — Import Libraries

Use:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Section 3 — Load Data

Use:

```
load_data()
```

Section 4 — Basic Data Understanding

Show:

```
shape
head()
tail()
info()
describe()
columns
dtypes
```

Section 5 — Data Quality

Check:

```
missing values
duplicates
unique values
invalid values
```

Section 6 — Data Cleaning

Use the standardized cleaning functions.

Section 7 — Derived Columns

Create:

```
delivery_days
delay_days
cost_per_km
total_logistics_cost
on_time_flag
```

Section 8 — EDA

Perform:

```
Univariate
Bivariate
Multivariate
```

Section 9 — KPI Analysis

Calculate all required KPIs.

Section 10 — Business Analysis

Answer the required business questions.

Section 11 — Visualization

Create required charts.

Section 12 — Business Insights

Write findings in simple business language.

Section 13 — Recommendations

Convert important findings into recommendations.

Section 14 — Final Conclusion

Summarize the project.

---

29. BUSINESS ANALYSIS TEMPLATE

For every business question, students should use this format:

```
BUSINESS QUESTION
-----------------
Which warehouse has the highest delay percentage?

DATA REQUIRED
-------------
warehouse
delivery_status
delay_days

ANALYSIS
--------
Group data by warehouse and calculate delay percentage.

VISUALIZATION
-------------
Bar chart

FINDING
-------
Write the actual finding based on the data.

BUSINESS MEANING
----------------
Explain why the finding matters.

RECOMMENDATION
--------------
Suggest a reasonable business action.
```

This structure is mandatory for important questions.

---

30. BUSINESS INSIGHT FORMAT

Students must NOT write only:

Warehouse A has 25% delays.

They should write:

```
Finding:
Warehouse A has the highest delay percentage among the
warehouses analyzed.
```

```
Business Meaning:
This indicates that Warehouse A may require further
investigation into processing and dispatch operations.

Recommendation:
The operations team should review processing time,
staff workload, dispatch procedures, and shipment handoff
activities at Warehouse A.
```

The recommendation must be based on the available evidence.

---

31. STATISTICAL ANALYSIS

The project should include beginner/intermediate statistical analysis.

Central Tendency

```
Mean
Median
Mode
```

Use examples such as:

```
Average delivery time
Median shipping cost
Most common shipping mode
```

Dispersion

```
Variance
Standard Deviation
```

Use examples such as:

```
Variation in delivery time
Variation in shipping cost
```

Correlation

Analyze relationships such as:

```
Distance vs Delivery Days
Distance vs Shipping Cost
Weight vs Shipping Cost
Processing Hours vs Delay Days
```

Use:

```
df.corr(numeric_only=True)
```

and Seaborn heatmaps where appropriate.

---

32. STREAMLIT APPLICATION

File:

```
app/streamlit_app.py
```

The Streamlit application is the final interactive business dashboard.

Dashboard Pages

Keep the application simple.

Page 1 — Executive Overview

Display:

```
Total Orders
Delivered Orders
Delayed Orders
On-Time %
Average Delivery Days
Total Logistics Cost
Average Logistics Cost
```

Page 2 — Delivery Analysis

Display:

```
Delivery status
Monthly delivery trend
Delay analysis
```

Page 3 — Route Analysis

Display:

```
Top routes
Route delays
Route costs
```

Page 4 — Warehouse Analysis

Display:

```
Orders by warehouse
Processing time
Delay percentage
```

Page 5 — Cost Analysis

Display:

```
Total cost
Average cost
Cost by mode
Cost by route
Cost trend
```

Page 6 — Partner Analysis

Display:

```
Partner shipments
Partner delay %
Partner cost
Partner damage rate
```

---

33. STREAMLIT REQUIRED FUNCTIONS

The Streamlit file must use these function names:

```
load_dashboard_data()

show_sidebar_filters()

show_overview()

show_delivery_analysis()
```

```
show_route_analysis()

show_warehouse_analysis()

show_cost_analysis()

show_partner_analysis()

main()
```

Each function should have a docstring.

---

34. STREAMLIT FILTERS

The sidebar should provide simple filters:

```
Warehouse
Origin City
Destination City
Shipping Mode
Delivery Partner
Delivery Status
```

If date filtering is implemented, use:

```
Order Date
```

Filters should update the displayed KPIs and charts.

---

35. STREAMLIT FLOW

```
Start Application
       ↓
Load Cleaned Data
       ↓
Show Sidebar Filters
       ↓
User Selects Filters
       ↓
Filter Data
       ↓
Calculate KPIs
       ↓
Display Charts
       ↓
Display Business Information
```

Run:

```
streamlit run app/streamlit_app.py
```

---

36. TESTING REQUIREMENTS

File:

```
tests/test_pipeline.py
```

Testing is intentionally simple.

Students should test:

- 1. Data loading
- 2. Data cleaning

- 3. Duplicate removal
- 4. Derived columns
- 5. KPI calculations
- 6. Output file creation

Required test function names:

```
test_load_data()

test_remove_duplicates()

test_create_derived_columns()

test_clean_data()

test_calculate_total_orders()

test_calculate_on_time_percentage()

test_save_cleaned_data()
```

---

37. REQUIREMENTS.TXT

The project should contain:

```
pandas
numpy
matplotlib
seaborn
jupyter
streamlit
pytest
```

Students may pin versions if required.

---

38. README REQUIREMENTS

The README must contain:

```
Project Title

Project Description

Business Problem

Objectives

Technology Stack

Project Structure

Installation

How to Run Cleaning Pipeline

How to Run Jupyter Notebook

How to Run Streamlit

Business Questions

KPIs

Key Insights

Recommendations

Future Improvements
```

---

39. FINAL BUSINESS REPORT

File:

```
reports/business_report.md
```

The report must contain:

39.1 Executive Summary

Short explanation of the project.

39.2 Business Problem

What problem was given by the client?

39.3 Data Summary

What data was analyzed?

39.4 Data Quality

What problems were found?

39.5 Cleaning Performed

What cleaning steps were applied?

39.6 KPI Summary

Show important KPIs.

39.7 Key Findings

List major findings.

39.8 Business Recommendations

List practical recommendations.

39.9 Conclusion

Summarize the analytical outcome.

---

40. COMPLETE PROJECT FLOW

The student must understand this complete flow:

```
CLIENT
  ■
```

```
  ▼
BUSINESS REQUIREMENT
  ■
  ▼
BRD
  ■
  ▼
FRD
  ■
  ▼
RAW DATA
  ■
  ▼
DATA UNDERSTANDING
  ■
  ▼
DATA QUALITY CHECK
  ■
  ▼
DATA CLEANING
  ■
  ▼
CLEANING PIPELINE
  ■
  ▼
CLEANED DATA
  ■
  ▼
EDA
  ■
  ■■■ Univariate
  ■■■ Bivariate
  ■■■ Multivariate
  ■
  ▼
BUSINESS ANALYSIS
  ■
  ■■■ Delivery
  ■■■ Delay
  ■■■ Route
  ■■■ Warehouse
  ■■■ Cost
  ■■■ Partner
  ■
  ▼
KPI ANALYSIS
  ■
  ▼
STATISTICAL ANALYSIS
  ■
  ▼
VISUALIZATION
  ■
  ▼
BUSINESS INSIGHTS
  ■
  ▼
BUSINESS RECOMMENDATIONS
  ■
  ▼
STREAMLIT DASHBOARD
  ■
  ▼
FINAL BUSINESS REPORT
```

---

41. FUNCTION MAP

This is the master function list that every student must follow.

src/data_loader.py

```
check_file_exists()
load_data()
```

src/data_cleaner.py

```
handle_missing_values()
remove_duplicates()
```

```
clean_text_columns()
clean_date_columns()
validate_numeric_columns()
validate_business_rules()
create_derived_columns()
clean_data()
save_cleaned_data()
```

src/business_analysis.py

```
calculate_total_orders()
calculate_total_delivered()
calculate_total_delayed()
calculate_on_time_percentage()
calculate_average_delivery_days()
calculate_average_delay_days()

calculate_total_shipping_cost()
calculate_average_shipping_cost()
calculate_total_fuel_cost()
calculate_total_logistics_cost()
calculate_average_logistics_cost()
calculate_average_cost_per_km()

analyze_route_performance()
analyze_route_cost()
analyze_route_delays()

analyze_warehouse_performance()
analyze_warehouse_delays()
analyze_warehouse_processing_time()

analyze_shipping_mode()
analyze_shipping_mode_cost()
analyze_shipping_mode_delays()

analyze_delivery_partner()
analyze_partner_cost()
analyze_partner_delays()
analyze_partner_damage_rate()

calculate_average_customer_rating()
calculate_return_rate()
calculate_damage_rate()
```

src/visualization.py

```
plot_delivery_status()
plot_monthly_orders()
plot_monthly_delivery_performance()
plot_delay_by_warehouse()
plot_delay_by_shipping_mode()
plot_delay_by_route()
plot_delay_distribution()
plot_cost_by_shipping_mode()
plot_cost_by_route()
plot_cost_trend()
plot_orders_by_warehouse()
plot_processing_time_by_warehouse()
plot_shipments_by_partner()
plot_partner_delivery_performance()
plot_partner_cost()
plot_customer_rating_distribution()
plot_rating_by_delivery_status()
```

app/streamlit_app.py

```
load_dashboard_data()
show_sidebar_filters()
show_overview()
show_delivery_analysis()
show_route_analysis()
show_warehouse_analysis()
show_cost_analysis()
show_partner_analysis()
main()
```

run_pipeline.py

```
run_pipeline()
```

---

42. FUNCTION RESPONSIBILITY RULE

Students must follow one important software design rule:

One function should have one clear responsibility.

Bad approach:

```
One function:
load data
clean data
calculate KPIs
create charts
save files
```

Better approach:

```
load_data()
      ↓
clean_data()
      ↓
calculate_kpis()
      ↓
create_chart()
      ↓
save_output()
```

This makes the project easier to understand, test, and maintain.

---

43. CODING STYLE RULES

Every function must have a docstring.

Example:

```
def calculate_total_orders(df):
    """Return the number of unique orders in the dataset."""
```

Use meaningful variable names.

Good:

```
total_orders
average_delivery_days
delay_percentage
```

Avoid:

```
x
y
a1
temp
abc
```

unless they are short-lived loop variables.

---

44. NOTEBOOK RULES

The notebook must contain Markdown explanations before important code.

Example:

```
## Delivery Performance Analysis

The purpose of this analysis is to understand how many
shipments were delivered on time and how many were delayed.
```

Then the code.

Then explain the result.

```
### Business Finding

The analysis shows that delayed shipments represent a
significant portion of the analyzed orders.
```

The notebook must tell a story.

---

45. ANALYSIS RULE

Students must follow:

```
QUESTION
   ↓
DATA
   ↓
CALCULATION
   ↓
VISUALIZATION
   ↓
FINDING
   ↓
BUSINESS MEANING
   ↓
RECOMMENDATION
```

Do not create random charts without a business purpose.

---

46. FINAL DELIVERABLES

Every student must submit:

```
logistics_delivery_analytics/
■
■■■ data/
■   ■■■ raw/
■   ■   ■■■ logistics_data.csv
■   ■■■ cleaned/
■       ■■■ logistics_cleaned.csv
■
■■■ notebooks/
■   ■■■ logistics_analysis.ipynb
■
■■■ src/
■   ■■■ __init__.py
■   ■■■ data_loader.py
■   ■■■ data_cleaner.py
■   ■■■ business_analysis.py
■   ■■■ visualization.py
■
■■■ app/
■   ■■■ streamlit_app.py
■
■■■ reports/
■   ■■■ business_report.md
■
■■■ tests/
■   ■■■ test_pipeline.py
■
■■■ run_pipeline.py
■■■ requirements.txt
■■■ README.md
■■■ .gitignore
```

---

47. ACCEPTANCE CRITERIA

The assignment is considered complete only when:

- Raw data is available.
- Cleaning pipeline runs successfully.
- Cleaned CSV is generated.
- Required columns are validated.
- Required derived columns are created.
- Jupyter notebook runs from beginning to end.
- EDA is completed.
- Required business questions are answered.
- Required KPIs are calculated.
- Required visualizations are created.
- Business insights are documented.
- Recommendations are documented.
- Streamlit application runs.
- Streamlit filters work.
- Basic tests pass.
- README is complete.
- Business report is complete.
- Standard function names are used.

---

48. STUDENT SUBMISSION CHECKLIST

Before submission:

```
[ ] Project folder name is correct
[ ] Folder structure is correct
[ ] File names are correct
[ ] Function names are correct
[ ] All functions have docstrings
[ ] Raw data is preserved
[ ] Cleaning pipeline works
[ ] Cleaned dataset generated
[ ] No major missing-value problems remain
[ ] Duplicate handling completed
[ ] Date columns converted correctly
[ ] Numeric columns validated
[ ] Derived columns created
[ ] EDA completed
[ ] KPI calculations completed
[ ] Business questions answered
[ ] Required charts created
[ ] Insights written
[ ] Recommendations written
[ ] Streamlit dashboard works
[ ] Tests run successfully
[ ] README completed
[ ] Business report completed
```

---

49. WHAT THE STUDENT IS ACTUALLY LEARNING

This project is not only about Pandas.

The student learns:

```
Python
  ↓
Data Handling
  ↓
Data Cleaning
  ↓
Reusable Functions
  ↓
Pipeline Thinking
  ↓
EDA
  ↓
Statistics
  ↓
Visualization
  ↓
Business Questions
  ↓
KPI Thinking
  ↓
Business Insights
  ↓
Business Recommendations
  ↓
Dashboard Development
```

The most important learning outcome is:

A Data Analyst does not simply create charts. A Data Analyst uses data to answer business questions and communicate evidence clearly to decision makers.

---

50. FINAL PROJECT EXPECTATION

At the end of this assignment, a student should be able to explain the project like this:

"I worked on a logistics delivery analytics project. I started by understanding the client's business requirements and converted them into functional requirements. I loaded and profiled the raw logistics data, identified data-quality problems, and built a reusable Python cleaning pipeline. After generating the cleaned dataset, I performed exploratory data analysis using Pandas, Matplotlib, and Seaborn. I calculated delivery, delay, warehouse, route, partner, and cost KPIs. I then answered business questions using data analysis and visualizations, documented business insights and recommendations, and finally built a Streamlit dashboard so business users could interactively analyze the results."

That is the expected project story.

---

51. PROJECT SUCCESS DEFINITION

The project is successful when a business user can answer:

```
How are our deliveries performing?

Where are delays occurring?

Which routes need attention?

Which warehouses need investigation?

How much are we spending?

Which transportation modes cost more?

How are delivery partners performing?

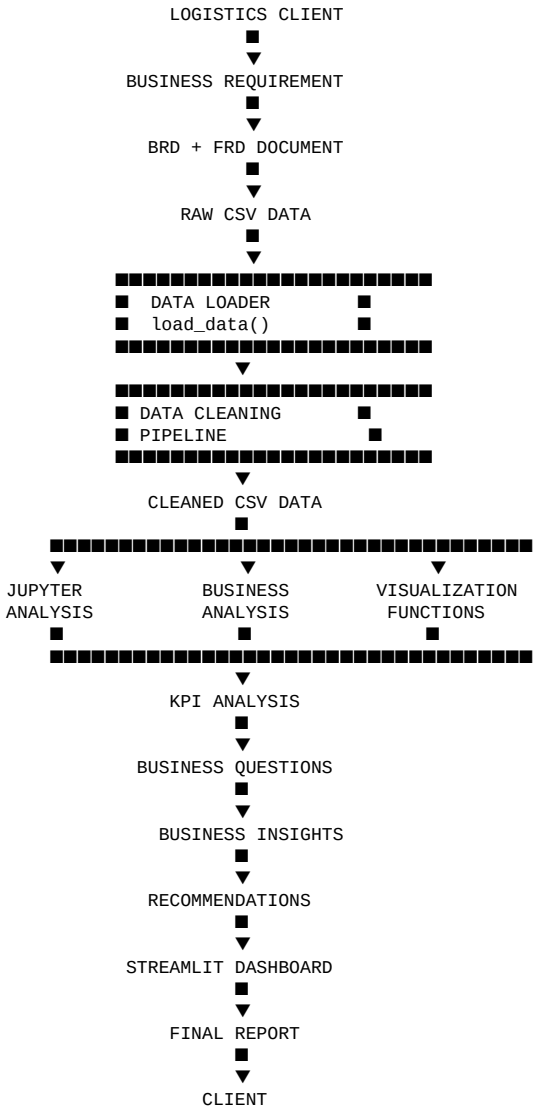
What customer-experience issues are visible?

What does the data tell us?
What should the business investigate or improve?
```

using the student's analysis and dashboard.

---

52. FINAL ARCHITECTURE



---

53. CORE RULE FOR THIS ASSIGNMENT

**Same problem.

Same folder structure.

Same file names.

Same function names.

Same required KPIs.

Same required business questions.

Same minimum visualizations.**

Students are free to write their own implementation inside the functions and explain their findings from the actual data, but the project interface and structure should remain standardized.

---

54. END OF ASSIGNMENT SPECIFICATION

Project: Logistics Delivery Performance & Cost Analytics

Role: Data Analyst / Business Analyst

Level: Beginner → Intermediate

Focus: Data Analytics + Business Analytics

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Streamlit

ML: Not included